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Subject code & Name :	ITUE301: Advanced Web Development Frameworks	Practical :	7	Academic Year :	2026-27

Task - 7

Aim

To implement secure token-based user authentication using JSON Web Tokens (JWT) and bcryptjs password hashing, constructing an Express authentication pipeline with registration, login, and custom authentication middleware to protect private API endpoints against unauthorized access.

Code

GitHub Link / Repository Link: https://github.com/Dharmikvaghani/AWDF-PRACTICAL.git

Key Implementation Highlights:

1. User model password encryption using bcryptjs salt rounds in pre-save middleware.
2. JWT token issuance upon registration/login with 7-day expiration.
3. authMiddleware verifying Authorization: Bearer <token> headers and attaching user context.

```
// middleware/auth.js
export const authenticate = (req, res, next) => {
  const authHeader = req.headers.authorization;
  if (!authHeader?.startsWith('Bearer ')) {
    return res.status(401).json({ error: 'Unauthorized: Missing token' });
  }
  const token = authHeader.split(' ')[1];
  try {
    req.user = jwt.verify(token, process.env.JWT_SECRET);
    next();
  } catch (err) {
    res.status(401).json({ error: 'Unauthorized: Invalid token' });
  }
};
```

Output

Output

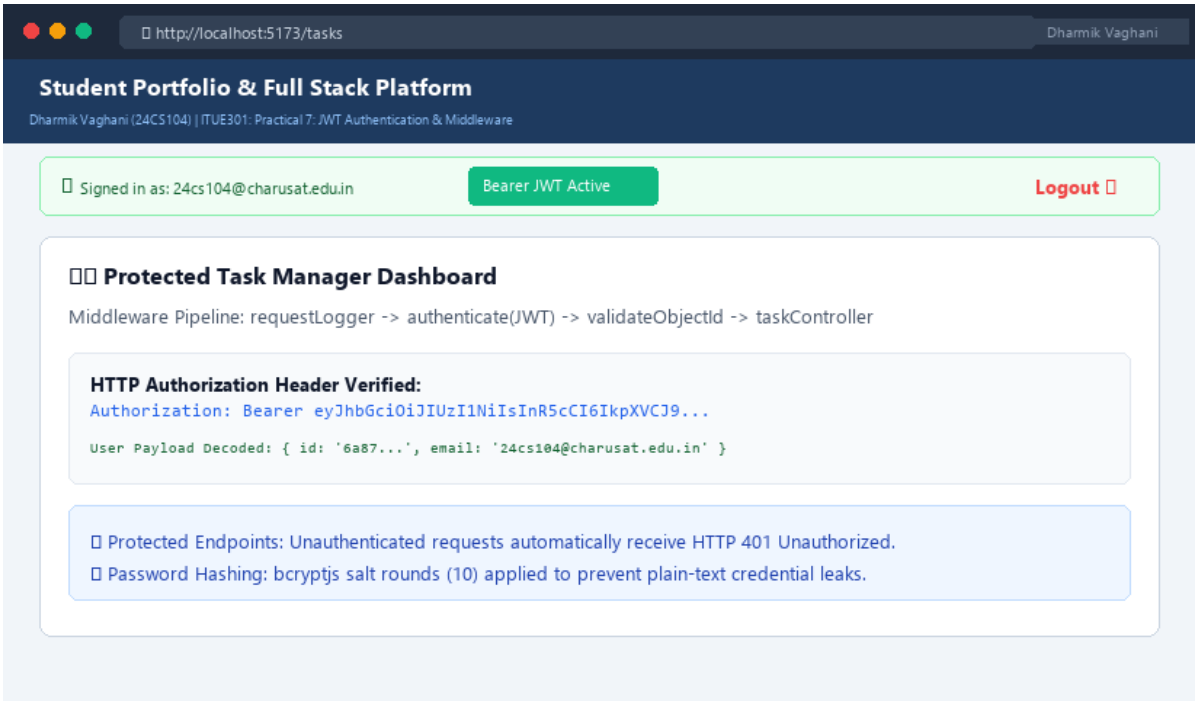


Figure 7: Execution Output & Functional Verification for Practical 7